Rules Win at Games KEY

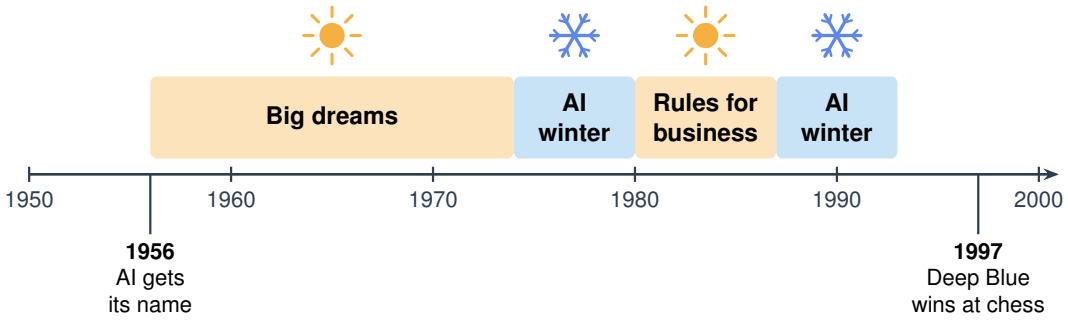
Name _____

Work with a partner. Compare answers when you finish each problem.

1. Did it come true? Did each promise come true by the time they said?

	Year	The promise	Came true by then?	If it came true later: when?
A	1958	A computer will soon walk, talk, see, write, make copies of itself, and know that it exists.	no	not yet
B	1958	Within 10 years, a computer will be the world chess champion.	no	1997
C	1965	Within 20 years, machines will do any work a person can do.	no	people disagree
D	1970	In 3 to 8 years, a machine will be as smart as an average person.	no	people disagree

2. Dreams and winters.



In 1956, AI got its name. When the money for AI stopped, people called it an AI winter.

Big promises, more than the facts show, are called hype.

When I hear a big promise about AI, I ask: Who says this? What do they get if I believe it?

3. Deep Blue.

In 1997, IBM's Deep Blue beat the world chess champion, Garry Kasparov.

Deep Blue checked about 200 million chess positions every second.

Did Deep Blue learn from examples? No. It used rules that people wrote, and speed.

4. Beat the rule bot. The bot checks its rules in order. It uses the first rule it can.

1	**Win**	If I have two in a row, I finish the line.
2	**Block**	If you have two in a row, I stop you.
3	**Fork**	I make two ways to win at the same time.
4	**Block a fork**	I stop you from making two ways to win.
5	**Center**	I take the middle square.
6	**Opposite corner**	If you are in a corner, I take the corner across from you.
7	**Corner**	I take an empty corner.
8	**Side**	I take an empty side square.

With all the rules on, I won _____ times, the bot won _____ times, and we tied _____ times.

The rule I turned off	**Did I beat the bot?**	**How did I win?**
Block	yes	I made two in a row. The bot did not stop me.

The rules the bot needs most: Win, Block, Block a fork, Center, Corner

5. Talk about it. Is Deep Blue smart?

I think Deep Blue **is** / **is not** smart, because Answers vary.

✂ *Cut out the cards. Add them to the class timeline.*

1956

Scientists meet at Dartmouth College. They name a new field: **artificial intelligence**.

Draw a picture here.

1997

IBM's computer **Deep Blue** beats the world chess champion, Garry Kasparov.

Draw a picture here.